ELL8299 Assignment Report

[HuggingFace Hub Model Link]

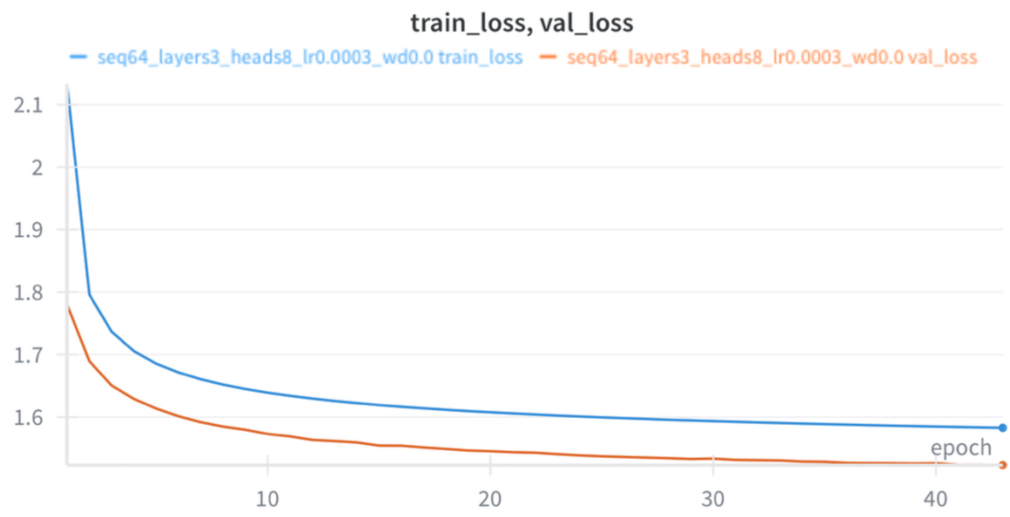
Part 1

Training

First, I ran the recommended base line configuration, as mentioned in the assignment instructions.

To be specific, the configuration was
Context_length: 64
Num_layers: 3
Num_attention_heads : 8
Hidden_dim: 300

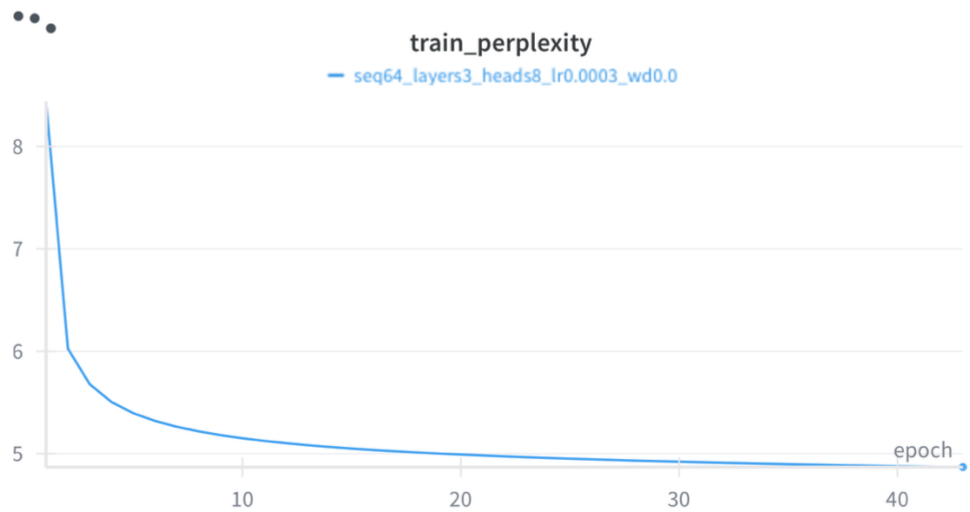
The train and validation loss curves are shown below.



1:Train/Val loss curve for baseline config

The final epoch train loss was: **1.58296**
The final epoch val loss was: **1.52328**

The train perplexity curve for the baseline model looks as follows:



2Train Perplexity over epochs for baseline

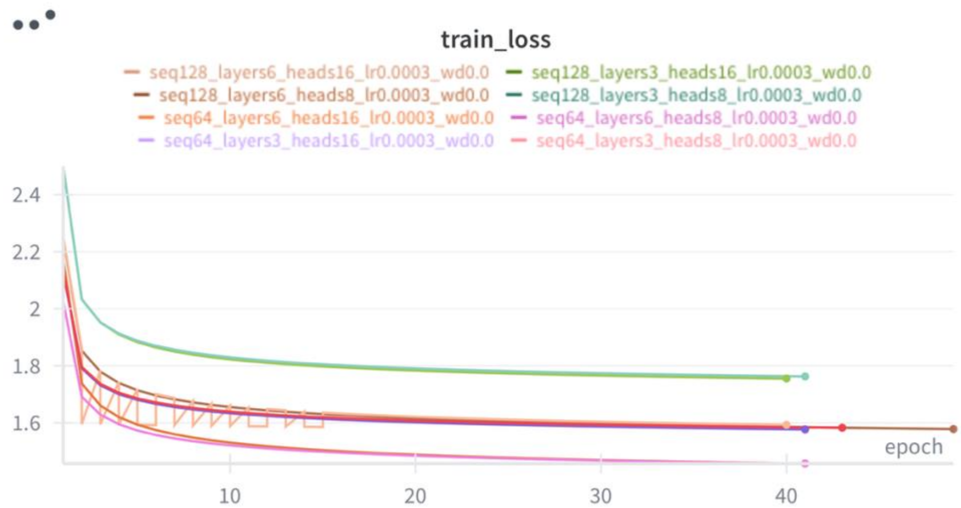
The final epoch train perplexity was: 4.86933

To experiment with multiple configurations, we ran a grid test over the following parameter choices:

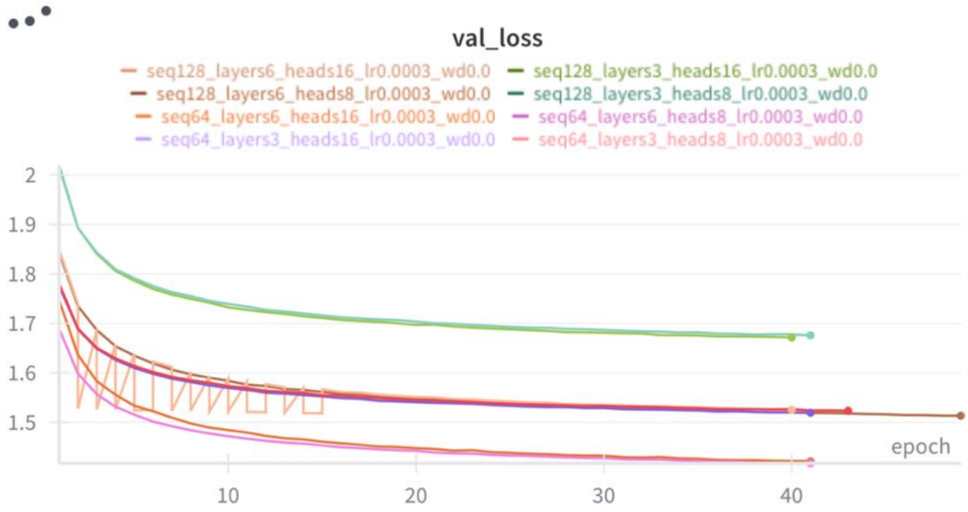
sequence_lengths=(64, 128)
num_layers=(3, 6)
num_heads=(8 16)

which resulted in a total of 8 unique configurations.

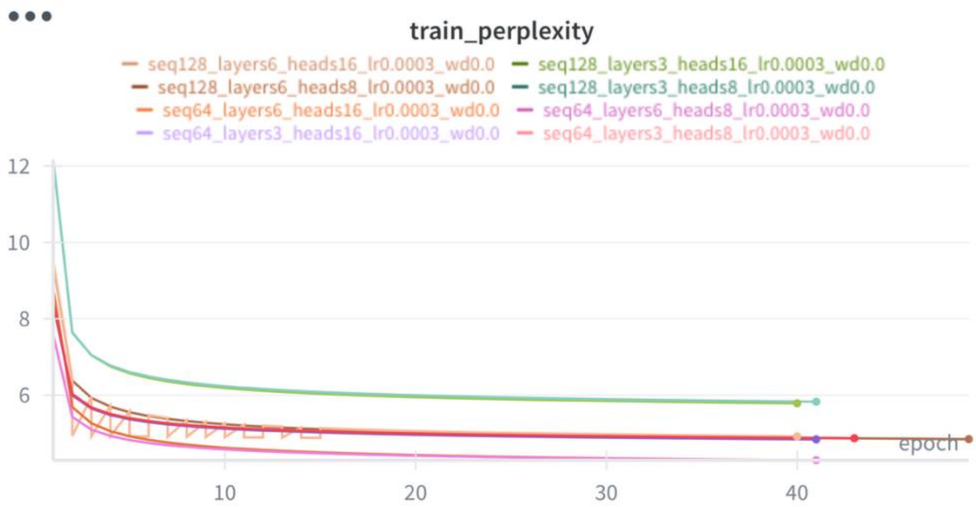
Below are are the train and val loss plots for all the configurations



3Train Loss over epochs for all configs



4Val Loss over epochs for all configs



5Train Perplexity over epochs for all configs

Upon examining these results, the model config with best metrics on the test sets was

Context_length: 64, layers:3, heads: 8

Metric	Metric Value
Test Loss	1.42533
Test BLEU	0.37189
Test Perplexity	4.15922

Inference

We select the model with above configuration to perform inference experiments.

Average Perplexity per token on samples from TinyStories validation set: ***1.56134***

Average BLEU score on 50 samples from TinyStories validation set: ***0.0133670***

Average time taken per token with Sampling/Greedy Decoding: ***0.00621***

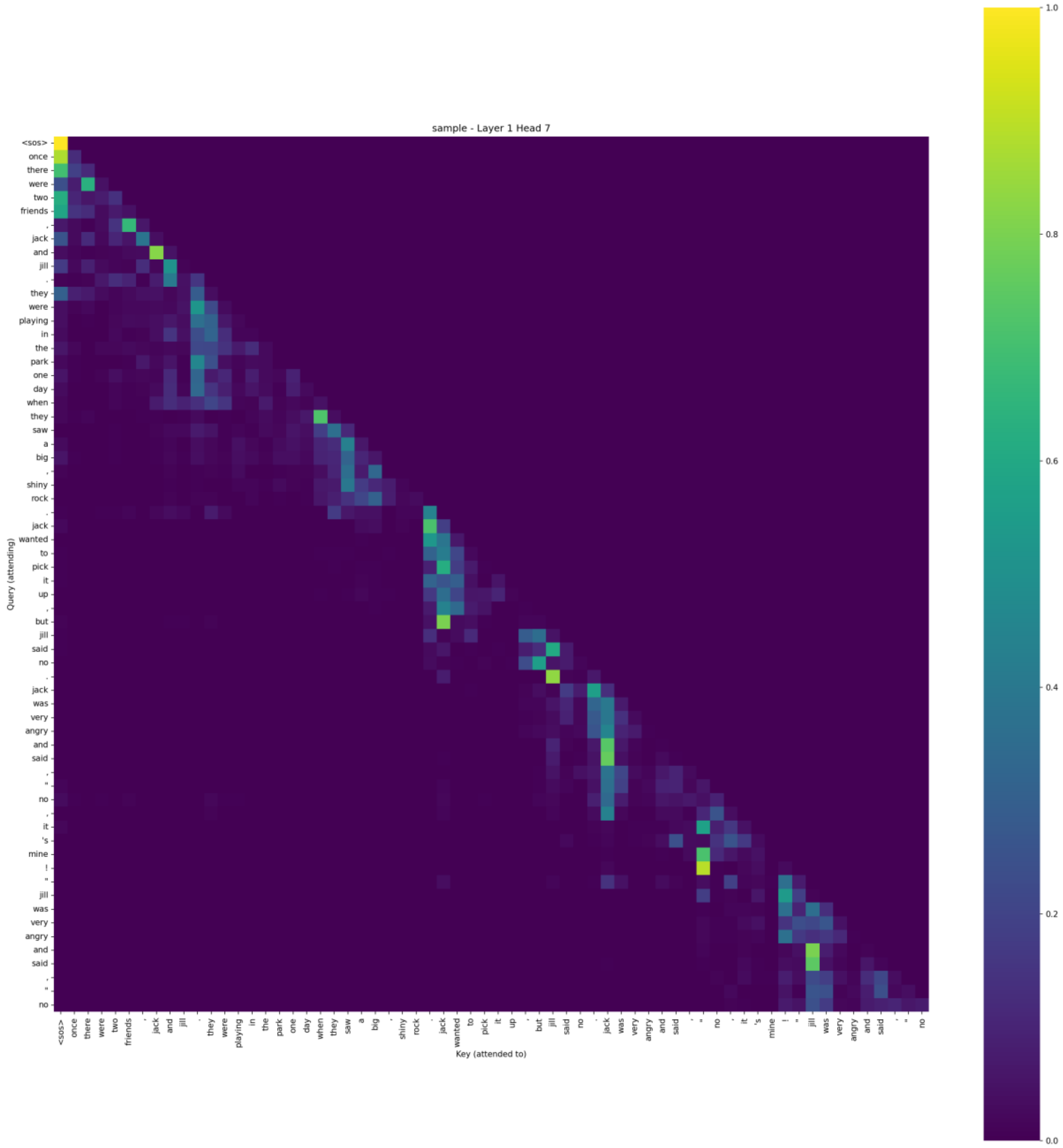
Below, some of the prompt/generated outputs and the relevant stats with them are showcased

Prompt: tom and mia were going
Generated Text: to the park with mom . they liked the park . it had swings , slides and a big sandbox . they put on their hats and coats and went to the park . they saw a big slide . it was very high and fast . they wanted to go on the slide . " mom ,
Perplexity per token: 1.1894
BLEU Score: 0.0010
Avg time per token: 0.0064 seconds

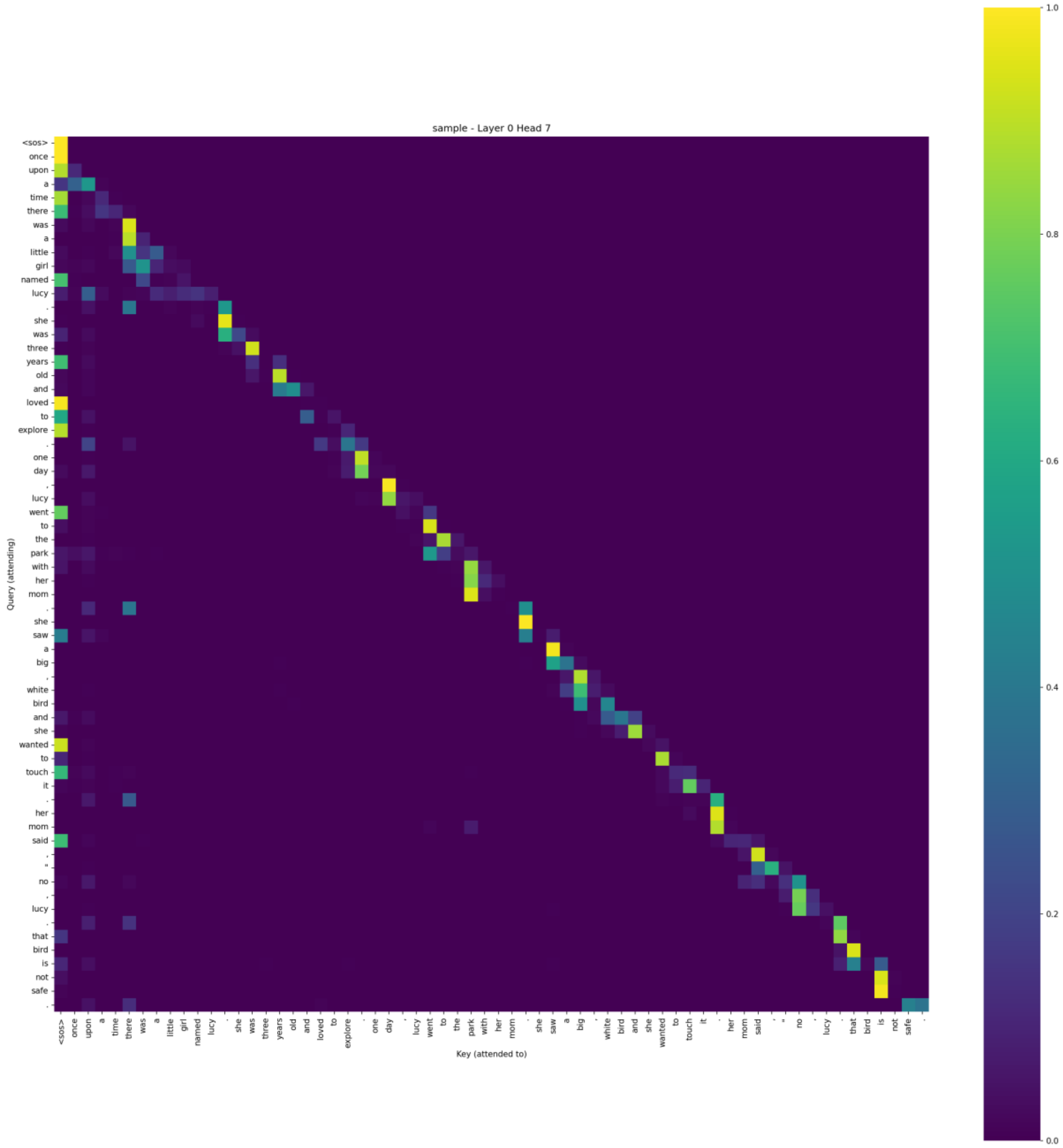
Prompt: one day , a small
Generated Text: boy named tim went to the park with his mom . they saw a big tree with a swing . tim wanted to play on the swing , but his mom said , " no , tim . you ca n't swing on the swing . " tim felt sad . he wanted to play on the swing
Perplexity per token: 1.2566
BLEU Score: 0.0
Avg time per token: 0.00605 seconds

The low BLEU scores can be explained by the high degree of difference between the ground truth text and the model generated completion.

<u>Attention heads:</u>



This particular head learns associations between nouns. For example: look at the attention scores associated with they tokens ‘jack’ ‘jill’.



This head has high attention scores around the punctuation tokens.

Part 2

<u>Beam Search</u>

k (Beam width)	Mean PPL / token	Mean BLEU	Mean Time per Token (seconds)
5	1.584	0.0223	0.012

10	1.606	0.012	0.022

Examples for k = 5

Prompt: jake loved to jog .
Generated Text: every morning , he would put on his shoes and go outside . he would put on his shoes and jog around the park . one day , jake was jogging in the park when he saw a big tree . he stopped to take a closer look . it was a big tree with lots of leaves
Perplexity per token: 1.937
BLEU Score: 0.0393
Avg time per token: 0.0149 seconds

Prompt: once upon a time ,
Generated Text: there was a little girl named lily . she loved to play outside and explore the world around her . one day , she went on a walk with her mommy and daddy . they walked and walked until they found a big tree with a hole in it . lily asked her mommy , " what 's
Perplexity per token: 1.434
BLEU Score: 0.0
Avg time per token: 0.0120 seconds

Examples for k = 10

Prompt: it was a sunny day
Generated Text: and jack was playing in the park . he was having so much fun ! suddenly , he heard a loud noise . he looked up and saw a big , dark cloud in the sky . it was a tornado ! jack was so scared that he started to run away . he ran as fast as
Perplexity per token: 1.9311
BLEU Score: 0.0068
Avg time per token: 0.025 seconds

Prompt: lila and ben were playing
Generated Text: in the snow . they made a big snowman with a hat and a scarf . they gave him a carrot nose and two rocks for eyes . they were very happy with their snowman . " look , our snowman is smiling ! " lila said . " yes , he is ! " ben said .

Perplexity per token: 1.206
BLEU Score: 0.0031
Avg time per token: 0.022 seconds

KV Caching

The table below shows the time to average time to generate one token calculate over 100 samples:

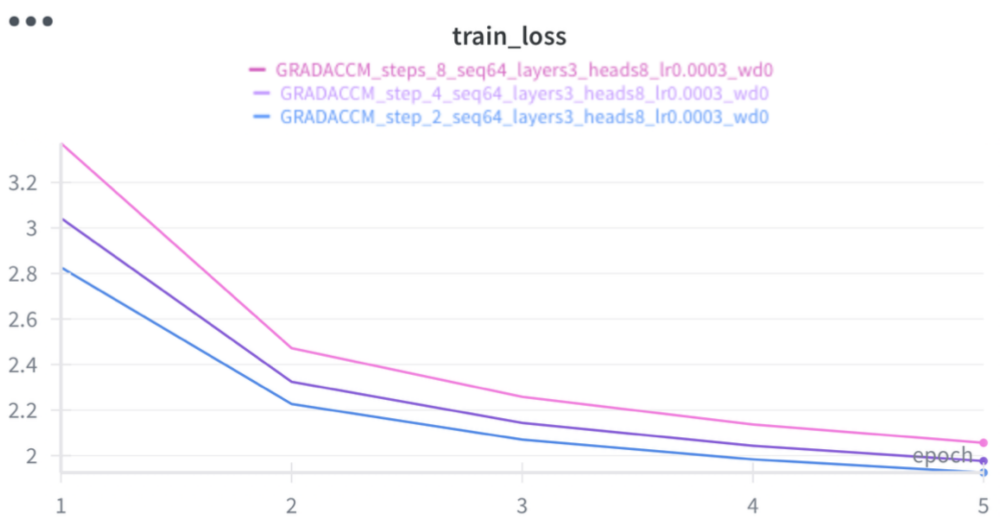
Without KV Caching	0.0097
With KV Caching	0.0059

Gradient Accumulation

For this part, we will train for a lower number of epochs to save on computation time, however the pattern are still clearly visible.

The image below shows the training loss curves for

Mini_batch_size = 16
Grad_accumulation_steps = 2, 4, 8



Gradient Accumulation Steps	Total Training Time	Total Epochs	Average Time/Epoch
2	11 minutes 14 seconds	5	2 mins 14 seconds
4	11 minutes 4 seconds	5	2 mins 12 seconds 800 ms
8	11 minutes 2 seconds	5	2 mins 12 seconds 400 ms

Gradient Checkpointing

	Without checkpointing	With checkpointing
Peak Memory Usage	5264.95 MB	4873.29 MB
Time Per Epoch	50.87 seconds	40.21 seconds